

LMPAN: A Lightweight Multi-Path Alignment Network for Joint Full-Duplex Acoustic Echo Cancellation and Noise Suppression

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1 MOTIVATION & OBJECTIVES

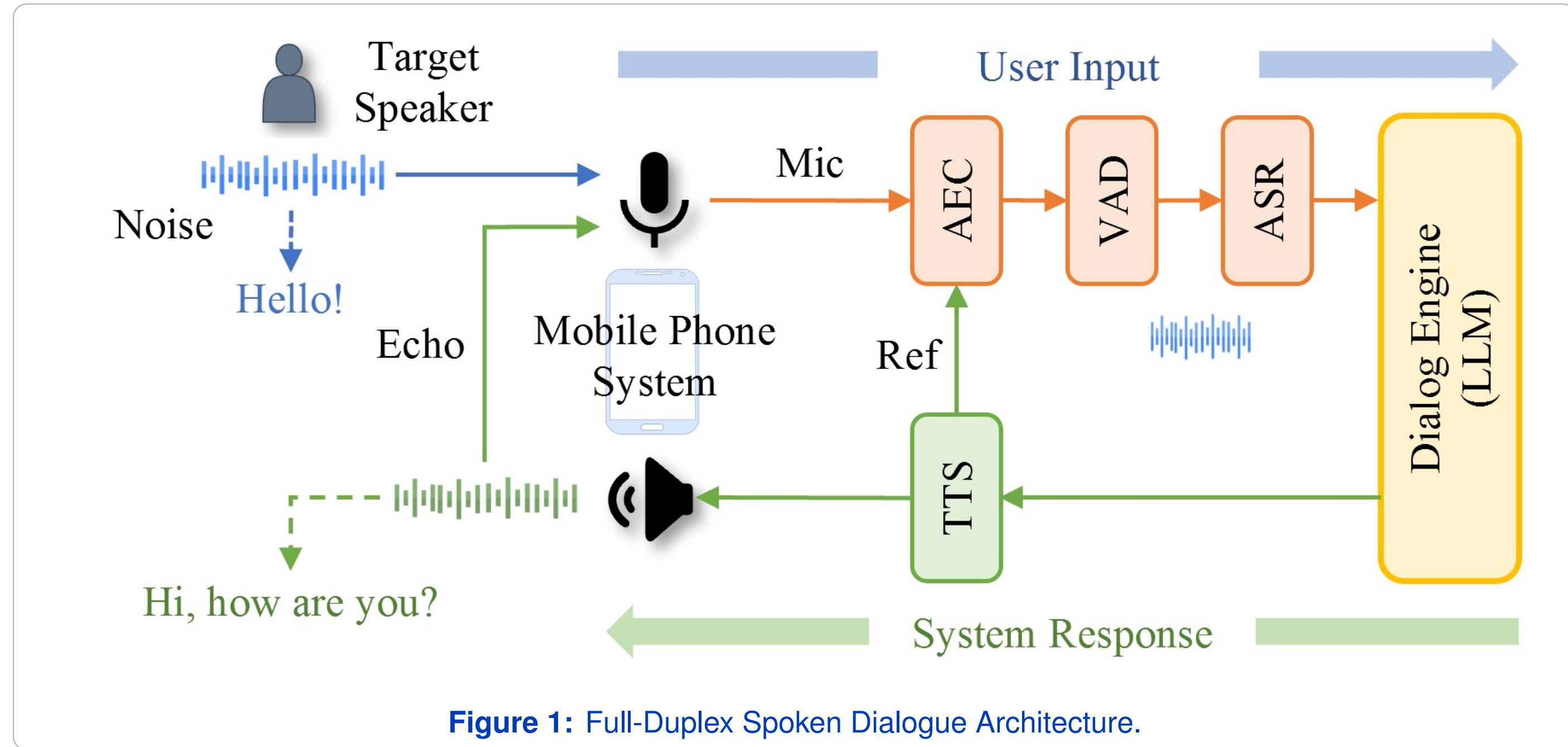


Figure 1: Full-Duplex Spoken Dialogue Architecture.

The Challenge:

- Joint AEC+NS on the edge:** full-duplex spoken dialogue on mobile devices demands joint acoustic echo cancellation (AEC) and noise suppression (NS) under a strict real-time budget.
- Reference-mic mismatch:** hardware distortion and dynamic acoustic conditions cause time-varying latency and energy mismatch between the far-end reference and microphone signals.
- Quality vs. latency:** lightweight models struggle to balance enhancement quality with real-time inference.

Our Contributions:

- Multi-path alignment:** pairwise alignment blocks correct time and energy mismatch across the ref/mic/LAEC paths.
- Attention fusion:** an attention mask dynamically integrates LAEC and microphone features, with no auxiliary VAD/DTD.
- Post-filtering + dynamic target:** controls residual interference to serve downstream ASR/VAD.
- Two-stage WavLM SSL training:** boosts perceptual quality via self-supervised embedding alignment.

2 PROPOSED ALGORITHM

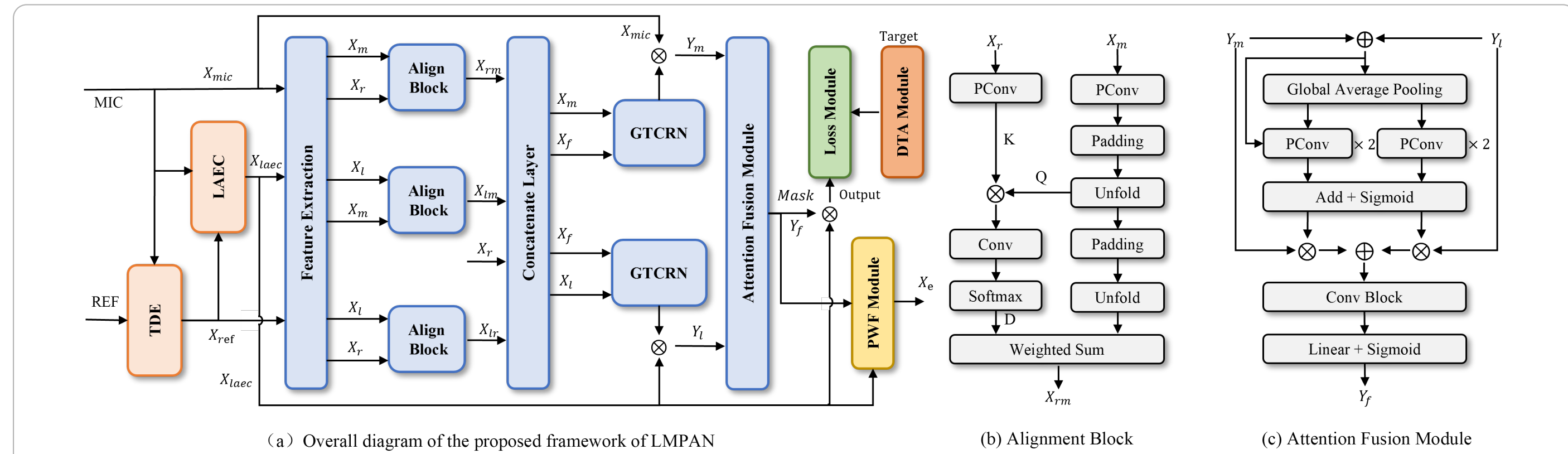


Figure 2: The LMPAN system. (a) Overall framework; (b) alignment block; (c) attention-based fusion.

A) Multi-Path Alignment

- Three pairwise align blocks:** ref-mic, mic-LAEC, ref-LAEC.
- Soft time-domain alignment:** a softmax over candidate delays $d \in [0, 100]$ frames (≈ 1 s) yields a soft delay distribution.
- Energy compensation:** learnable per-path energy scaling removes amplitude mismatch.

B) Attention-based Fusion

- Dual-stream refinement:** two GTCRN branches refine the LAEC and microphone streams; a multi-scale channel-attention mask M gates their combination:

$$Y_f = M \odot Y_l + (1 - M) \odot Y_m$$

(1)

- No auxiliary supervision:** requires no VAD / double-talk detector.

C) Post-filtering & Dynamic Target

- Residual scaling:** $\alpha = 0.4$ prevents over-suppression.
- Dynamic training target:** $t = s + \gamma n' + \beta e'$, where γ sets the target SNR and β the target SER — benefiting downstream ASR/VAD.

3 TWO-STAGE SSL TRAINING

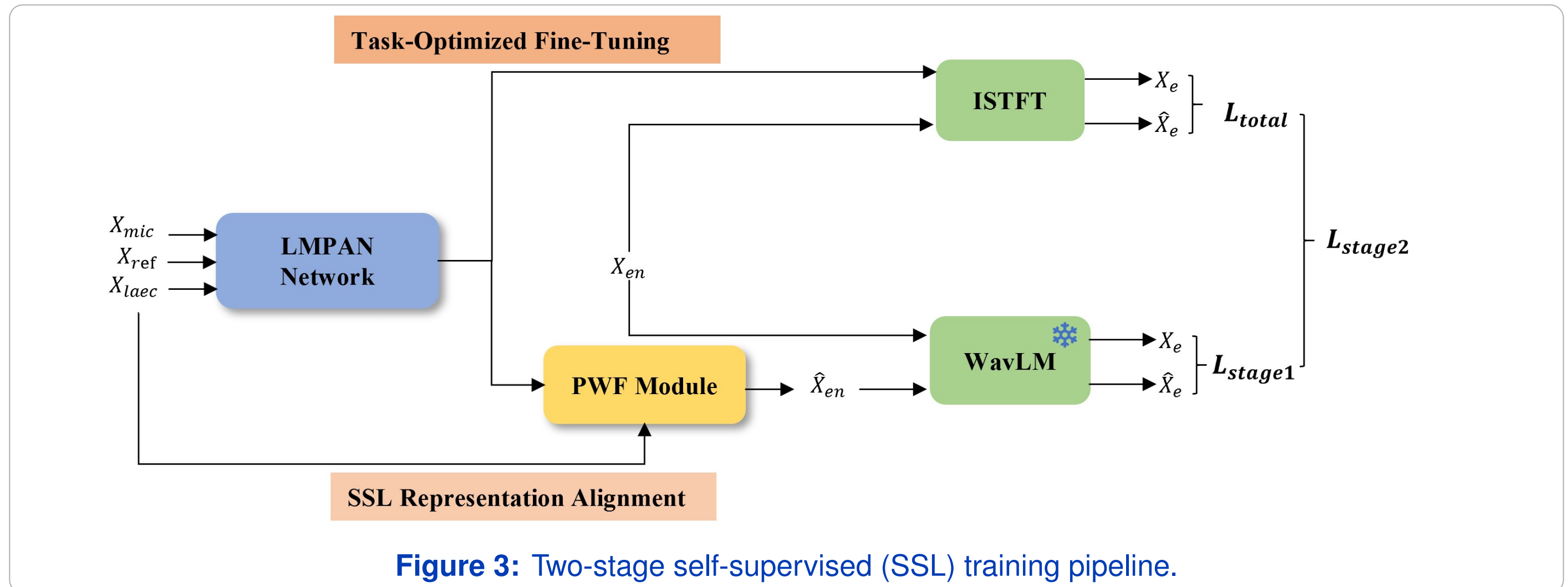


Figure 3: Two-stage self-supervised (SSL) training pipeline.

- Stage 1:** $\mathcal{L} = \mathcal{L}_{SSL}$ — freeze WavLM-Large and align embeddings with an MSE loss.
- Stage 2:** joint optimisation:

$$\mathcal{L} = 10 \mathcal{L}_{total} + 0.5 \mathcal{L}_{SSL} \quad (2)$$

$$\mathcal{L}_{total} = \mathcal{L}_{spec} + 0.1 \mathcal{L}_{echo} + 0.2 \mathcal{L}_{si-snr} + 0.8 \mathcal{L}_{pmsqe} \quad (3)$$

4 EXPERIMENTAL RESULTS

4.49	−9.87 pp	48.22 dB
MOS _{avg} (AEC Challenge 2023)	WER reduction (full pipeline)	ERLE (best echo suppression)

Table 1: AEC Challenge 2023 Blind Test

#	Method	Params	MACs	Double-Talk			ST-FE		MOS _{avg}
				EMOS	DMOS	ERLE	EMOS	DMOS	
Baselines									
–	DeepVQE (E2E)	0.82M	315M	4.62	4.02	65.7	4.61	4.36	4.40
–	Align-ULCNet (Hyb.)	0.69M	100M	4.60	3.80	–	4.77	4.28	4.36
–	TBNN (Hybrid)	9.56M	–	4.72	4.16	–	4.70	3.91	4.37
LMPAN (ours)									
1	Base (One-stage)	0.24M	65M	4.28	3.69	42.33	4.60	4.09	4.17
2	+MA	0.32M	82M	4.43	3.89	45.21	4.62	4.29	4.31
3	+MA+AFM	0.48M	126M	4.51	4.02	48.22	4.65	4.38	4.39
4	+MA+AFM+SSL-only	0.48M	126M	4.58	4.09	46.43	4.66	4.42	4.44
5	+MA+AFM+STL	0.48M	126M	4.63	4.17	47.15	4.71	4.44	4.49
6	+MA+AFM+STL+DTA	0.48M	126M	4.59	4.12	45.04	4.66	4.40	4.44

MA: multi-path alignment; AFM: attention fusion module; DTA: dynamic target adaptation; SSL-only: SSL loss only (no Stage-2); STL: two-stage training with SSL. Best in **bold**.

Table 2: Downstream Tasks on Real Double-Talk

Method	SER: [−15, −10] dB			SER: [−20, −15] dB		
	DCF ↓	WER ↓	TIR ↑	DCF ↓	WER ↓	TIR ↑
One-stage	4.68	11.08	90.96	9.38	24.25	85.17
+MA	3.42	9.57	91.53	7.22	21.57	87.96
+MA+AFM	2.55	8.36	92.82	5.85	19.38	89.17
+MA+AFM+SSL	2.35	8.16	93.12	5.45	18.58	90.17
+MA+AFM+STL	1.98	6.56	94.34	4.68	17.18	91.47
+MA+AFM+STL+DTA	1.55	4.34	92.54	3.75	14.38	93.85

VAD (DCF %), ASR (WER %), FSDS (TIR %) on real-world DT test set; SNR ∈ [5, 20] dB. Best in **bold**.

Table 3: Dynamic Target SER, Ablation

SER _r (dB)	WER ↓	PESQ ↑	ERLE (dB) ↑
20	13.12	2.22	35.49
25	10.24	2.35	42.23
30	11.34	2.39	44.56
35	11.55	2.34	45.11

Evaluated on simulated DT test set. SER_r=25 dB optimal for ASR; 30 dB for PESQ; 35 dB for ERLE.

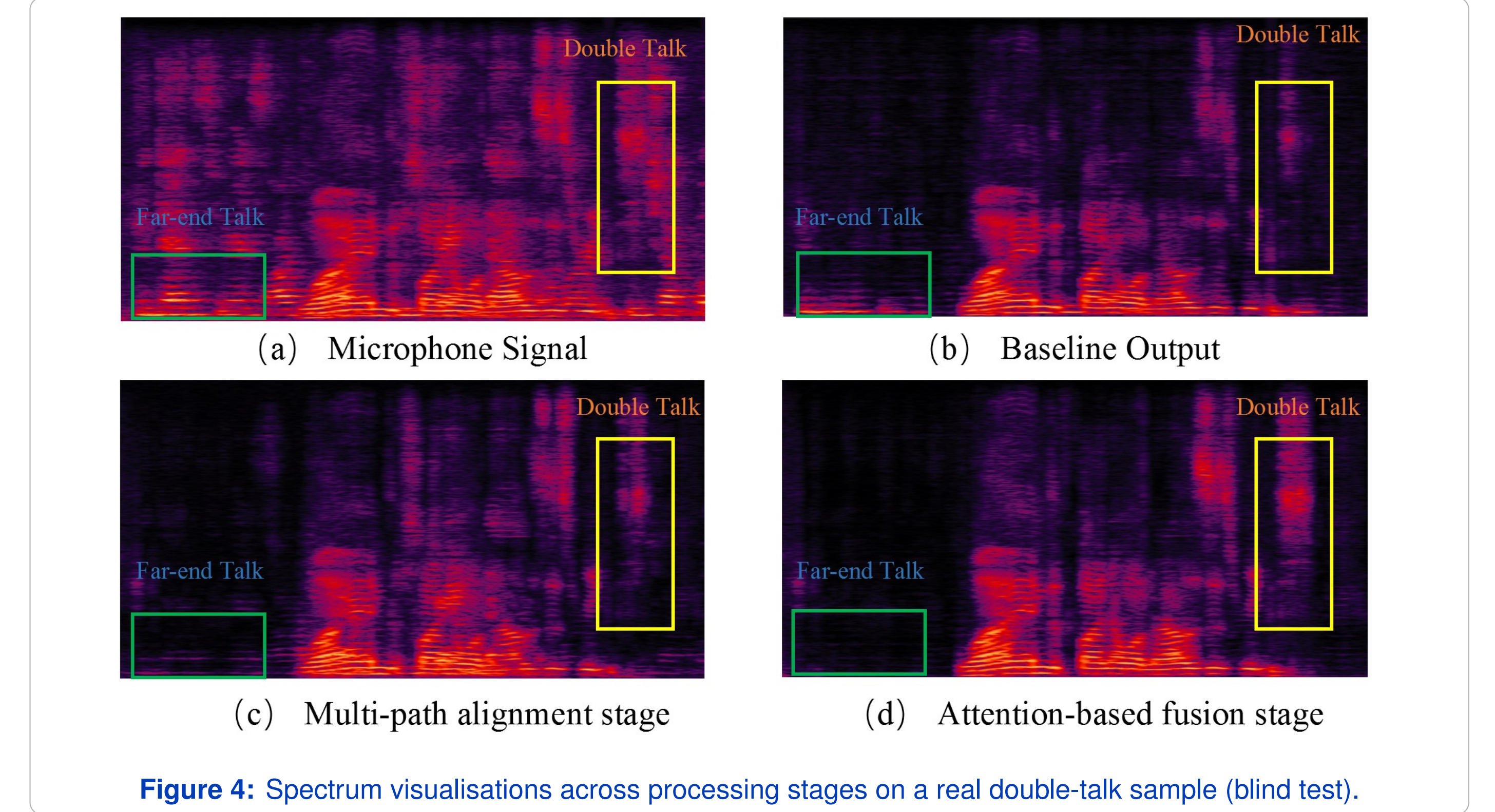


Figure 4: Spectrum visualisations across processing stages on a real double-talk sample (blind test).

Key Takeaways:

- Efficiency:** LMPAN surpasses SOTA lightweight models (DeepVQE, Align-ULCNet) at only 0.48M params / 126M MACs — 1.7× fewer parameters than DeepVQE.
- Progressive gains:** each module contributes incrementally: Base 4.17 → +MA 4.31 → +AFM 4.39 → +SSL 4.44 → +STL **4.49**.
- DTA for downstream:** task-related interference retention should be selected per downstream objective — SER_r=25 dB gives best WER (10.24); 35 dB maximises ERLE (45.11 dB).
- Full pipeline improvement:** in the hardest SER= [−20, −15] dB condition, DCF/WER/TIR improve by 5.63/9.87/8.68 pp over one-stage.

5 MAIN CONCLUSIONS

- Only **0.48M** parameters / **126M** MACs enable real-time inference on edge devices (RTF = 0.18 on ARM Cortex-A78).
- MOS_{avg} **4.49** surpasses the SOTA lightweight baseline (DeepVQE, 4.40) with ERLE up to **48.22 dB**.
- Multi-path alignment + attention fusion + two-stage SSL + DTA jointly guarantee speech integrity and downstream task performance.
- Dynamic target adaptation provides a tunable trade-off between enhancement quality and downstream ASR/VAD robustness.

6 REFERENCES

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